CryptoBot — Trading Strategy

A 24/7 systematic crypto scalper I built around a 4-layer decision brain, per-coin tuned parameters, and dip-aware entries. Long-only on live (andX), full long/short on the parallel sim.

Philosophy

This is a **systematic scalper**, not discretionary. Every decision — when to enter, how much to size, when to exit — comes out of a deterministic pipeline:

1. Scan the universe every 1–8s (depends on the active risk mode).
2. Score each candidate with 24 indicators + an ML ensemble.
3. Filter through the active mode's gate and the per-coin tier.
4. Size via Kelly + risk-parity, hard-capped by the mode's max position %.
5. Execute through andX (BTC/ETH/ANDX1 via the documented HMAC API) and in parallel through sim (full ~30-coin universe on Alpaca data).
6. Manage every open position every 1s — SL / TP / trail / harvest / fade / dump / emergency.

The whole system is **fee-aware** and **per-coin tuned**. No one-size-fits-all parameters. BTC harvest is +1.6%; PEPE harvest is +2.7%. Same formula, different ATR.

The 4-Layer Brain

Layer 1 — Signal (analysis.py + indicators_pro.py)

16 classic technical indicators plus 3 pro indicators. Each indicator outputs a directional score; the weighted sum is the **scalp_score** (0–100).

Layer 2 — ML brain (ml_engine.py + ml_ensemble.py)

XGBoost + LightGBM ensemble, calibrated to honest win-probability via isotonic regression. Trained on 2000+ historical trades. Walk-forward cross-validation, AUC ~0.81. Feeds Kelly sizing.

Layer 3 — Sizing (kelly_sizing.py)

Kelly criterion on the calibrated probability, multiplied by risk-parity (inverse-ATR), then by a sector-concentration penalty and a conviction multiplier. Always respects `max_single_position_pct` (mode hard cap — e.g. 10% for AGGRESSIVE).

Layer 4 — Execution (app.py)

Layers a market-regime + goal `size_boost` on top of Kelly. Falls back to the confidence-ladder if Kelly returns no-edge. Routes to docs API for BTC/ETH/ANDX1; everything else runs sim-only.

The 24 Indicators

Trend

- **SMA** — 10 / 20 / 50 / 100 / 200 period alignment
- **EMA** — 9 / 21 crossover (bull/bear cross detection)
- **Ichimoku Cloud** — Tenkan, Kijun, Senkou A/B, price-vs-cloud
- **SuperTrend** — ATR-based trailing line, flip detection
- **Trend structure** — higher-highs / higher-lows over recent swings

Momentum

- **RSI** — overbought/oversold + bullish/bearish divergence
- **MACD** — line, signal, histogram, crossover, momentum acceleration
- **Stochastic** — %K and %D, overbought/oversold + crosses
- **ADX** — trend strength (weak < 20 < trending < 40 < strong)
- **TTM Squeeze** — BB-inside-Keltner detection + squeeze momentum direction

Volume / Money Flow

- **OBV** — On-Balance Volume vs 20-bar average
- **CMF** — Chaikin Money Flow (buy vs sell pressure)
- **VWAP** — price above/below
- **Anchored VWAP** — anchored from the last swing low, % distance
- **Relative volume (rvol)** — current bar volume / 20-bar avg. Hard gate at rvol ≥ 0.5 (skip thin tape).

Volatility / Range

- **ATR** — Average True Range, drives SL/TP sizing
- **Bollinger Bands** — %B (position within bands), squeeze detection
- **52-week range** — % distance from high/low
- **BB Squeeze** — low-volatility coiling state

Structure / Levels

- **Support / Resistance** — pivot, R1/R2, S1/S2
- **Fibonacci** — 0 / 0.236 / 0.382 / 0.5 / 0.618 / 0.786 / 1.0 retracements
- **Swing highs / lows** — last 4 of each

Candlestick Patterns (24 detected)

Bullish: Hammer, Inverted Hammer, Bullish Engulfing, Morning Star, Three White Soldiers, Bullish Marubozu, Piercing, Bullish Harami, Tweezers Bottom, Three Inside Up, Bullish Belt Hold, Dragonfly Doji

Bearish: Hanging Man, Shooting Star, Bearish Engulfing, Evening Star, Three Black Crows, Bearish Marubozu, Dark Cloud, Bearish Harami, Tweezers Top, Three Inside Down, Bearish Belt Hold, Gravestone Doji

Multi-Timeframe

The top 3 candidates from the 5m scan get re-analyzed on 15m candles. Final score = 0.6 × 5m + 0.4 × 15m.

Risk Modes

The master knob. Every entry/exit threshold reads from the active mode. Switching is atomic — flip the dropdown and the next tick uses the new mode.

Param	CONSERVATIVE	REGULAR	AGGRESSIVE
Scan interval	8s	5s	1s
Min confidence	70	60	40
Scalp gate	35 (high)	22	10 (loose)
Max positions	3	6	20
Max size / position	20%	20%	10% (many small)
Entries per scan cycle	1	2	4–6
Min hold	90s	60s	10s
Harvest target	+1.5%	+2.0%	+1.2%
Re-entry cooldown	5 min	3 min	20s
Emergency stop	-4%	-6%	-8%
Use case	only A-grade setups	balanced default	constant trading across many coins

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DIP Detector (the long-only edge)

When DIP MODE is on, the bot only trades qualifying dips. A dip earns points across four pillars (total 100):

Pillar 1 — Oversold momentum (max 30 pts)

- RSI ≤ 28 (deeply oversold): 18 pts
- RSI ≤ 35 (oversold): 12 pts
- Fast RSI drop (8+ pts in a few bars): 8 pts
- Stochastic K ≤ 20: 8 pts
- BB %B ≤ 0.1 (at lower band): 6 pts

Pillar 2 — At support (max 25 pts)

Distance to EMA21 / SMA20 / S/R / S1 pivot / VWAP: - Right at support (≤ 0.3% above): 14 pts - Very near (≤ 0.8%): 10 pts - Near (≤ 1.5%): 6 pts - Hard skip: price has already broken below the level

Plus: CMF ≥ 0.05 (real money buying the dip): +5 pts

Pillar 3 — Reversal candle (max 20 pts)

- High-value reversals (Hammer, Bullish Engulfing, Morning Star, Piercing, Dragonfly Doji, Three White Soldiers): 12 pts
- Medium reversals (Tweezers Bottom, Bullish Harami, Three Inside Up): 7 pts
- Bullish RSI divergence: +8 pts

Pillar 4 — Higher timeframe still constructive (max 15 pts)

- Weekly trend BULLISH: 12 pts
- Weekly trend MIXED: 6 pts
- Hard penalty: Weekly BEARISH → −15 pts (falling-knife guard)
- Weekly RSI in 40–65 range: +3 pts
- Above EMA9 (bounce starting): +4 pts

Hard guards

- Price below daily SMA200 AND weekly bearish: skip outright (bear-market dip).

Threshold

Each coin has its own `dip_threshold` (55 for blue chips, 60 for majors, 70 for memes). A dip qualifies when score ≥ threshold. Qualified dips get a **+40 scalp_score boost** so they outrank generic trend signals.

Per-Coin Strategy Formula

Every coin's SL / TP / harvest is derived from three inputs:

```
INPUTS
  rt_fee             = round_trip_fee(coin)   # 0.5% BTC/ETH, 0% ANDX1
  atr_pct            = 75th percentile of (hi-lo)/close over 168h
  audit_max_loss     = worst historical -% for this coin (from trade history)

FORMULA
  edge_floor         = max(0.3%, 1.5 × rt_fee)
  sl_min             = max(0.3%, 0.5 × atr_pct)
  sl_max             = min(1.5 × atr_pct, audit_max_loss × 0.8, 6%)
  tp_min             = max(edge_floor, 0.8 × atr_pct)
  tp_max             = min(3.0 × atr_pct, 10%)
  harvest_threshold  = atr_pct + rt_fee   # net = 1 ATR after fees
  min_hold_seconds   = 3 (zero-fee) | 5 (calm) | 15 (medium) | 25 (meme)
  dip_threshold      = 55 (atr<1.5%) | 60 (atr<3%) | 70 (meme)
  scalp_gate         = 10 (zero-fee) | 14 (paying fees) +4 if win_rate<45%
```

Live output (auto-tuned on real ATR)

Coin	ATR	Audit	SL	TP	Harvest
BTC	1.08%	-16.4%	0.5–1.6%	0.9–3.2%	+1.58%
ETH	1.46%	-19.8%	0.7–2.2%	1.2–4.4%	+1.96%
SOL	1.72%	—	0.9–2.6%	1.4–5.2%	+2.22%
ADA	2.42%	—	1.2–3.6%	1.9–7.3%	+2.92%
SUSHI	2.57%	—	1.3–3.9%	2.1–7.7%	+3.07%
PEPE	2.15%	—	1.1–3.2%	1.7–6.4%	+2.65%
ANDX1	1.00%	—	0.5–1.5%	0.8–3.0%	+1.00%

Re-tuning is one click — the bot pulls fresh ATR + audit each time.

Position Sizing (Kelly + clamps)

Every entry runs through:

```
kelly_qty       = kelly_position_size(
                    ml_probability,         # calibrated win-prob from ensemble
                    technical_score,        # scalp_score
                    atr_pct,                # for risk-parity
                    sector_concentration,   # penalty if too much in one class
                    max_single_position_pct # mode hard cap
                  )
kelly_qty      *= mode.size_boost × goal_size_scale
deployed_usd    = kelly_qty × price
```

Goal-based throttling

- Daily target hit → size × 0.5, gate +25
- Near target (80%) → size × 0.8, gate +10
- Drawdown 30%+ from session peak → "guard mode" (A-grade signals only)
- Drawdown 40%+ → hard pause until reset

Exit Management (9 mechanisms, checked every 1s)

For every open position, in this order:

1. **Emergency stop** — gain ≤ -emergency_stop_pct (mode-specific). Bypasses min-hold.
2. **Min hold** — skip exit checks until per-coin min_hold_seconds elapses.
3. **Stop loss** — gain ≤ -SL_pct. Direction-aware (long below entry, short above).
4. **Take profit** — gain ≥ TP_pct. Direction-aware.
5. **Harvest** — gain ≥ harvest_threshold (per-coin). Locks small wins fast.
6. **Trail lock** — peak gain ≥ trail_lock_pct AND drop ≥ 0.5% from peak.
7. **Trail breakeven** — peak gain ≥ trail_be_pct AND gain back to 0.
8. **Fade protect** — peak ≥ fade_threshold AND gain dropped fade_drop % from peak (e.g. peaked at +3%, now at +1.8%).
9. **Dump bleed** — gain ≤ dump_bleed_pct (slow leak cut).
10. **Smart stop reassess** (sim only) — if SL hit but indicators still bullish, brief chance to recover.

Fee Awareness

```
fee_per_side(BTC/ETH) = 0.25%     # andX taker
fee_per_side(ANDX1)   = 0.00%
round_trip(coin)      = 2 × fee_per_side(coin)
```

On a $170 BTC trade: $0.85 round-trip fee. The harvest threshold's `atr_pct + rt_fee` design means the harvest target already includes the fee budget — the bot doesn't need explicit fee-subtraction logic if the formula's gross targets are trusted.

ANDX1 is the **frequency engine**. Zero fees mean tiny moves (+0.3%) are profitable, so it runs the tightest hold (3s) and gate (10).

Learning System (closes the loop)

After every trade, three records update:

- `ticker_memory.json` — per-coin win rate + average gain/loss. Future Kelly sizing on the same coin scales 0.4× to 1.1× by this reputation.
- `learning_data.json` — per-indicator-category accuracy. Each indicator's weight is multiplied by 0.85×–1.15× based on its hit rate over the last 50+ trades. Gentle (no flip-flop), only kicks in after 50+ trades.
- `trade_history.json` — full feature vector + outcome. Retrains the XGBoost+LightGBM ensemble every 10 trades.

Architectural Choices

- **5m candles** — best signal:noise ratio for scalping. 1m has too much noise, 15m too slow for active recycling.
- **Kelly over fixed %** — high-confidence trades get bigger, low-confidence trades get smaller. Hard-capped to avoid Kelly's classic "go all-in on +EV" failure mode.
- **ML on top of TA** — the 24 indicators have known false-positive patterns (e.g. RSI divergence in trending markets). The ensemble learns which combinations have predicted wins historically.
- **Long-only live** — andX's documented API supports spot longs only on BTC/ETH/ANDX1. Sim runs long+short across ~30 Alpaca coins for comparison.
- **Per-coin strategies** — BTC's 0.5% range gets stopped out instantly by SOL's 1.7% noise. One set of SL/TP across all coins either wastes opportunities on calm coins or churns on volatile ones.

Files

File	Purpose
`app.py`	Main engine, Flask server, trading loops
`analysis.py`	16 classic indicators + scoring
`indicators_pro.py`	TTM Squeeze, SuperTrend, Anchored VWAP
`dip_detector.py`	4-pillar dip scoring
`coin_strategies.py`	Per-coin tiers + auto-tuner
`kelly_sizing.py`	Kelly + risk-parity + sector penalty
`ml_engine.py` / `ml_ensemble.py`	ML brain
`learner.py`	Adaptive weights + ticker memory
`andx_client.py`	andX exchange adapter (HMAC)
`portfolio_live.py`	Live portfolio mirror
`portfolio_sim.py`	Paper portfolio
`hybrid_client.py`	Alpaca data + andX execution
`screener.py`	Universe selection